



Gefördert durch:



aufgrund eines Beschlusses  
des Deutschen Bundestages

# FEATURES OF EVENT-DRIVEN MESSAGE QUEUEING ARCHITECTURES IN MANUFACTURING – A REFERENCE MODEL FOR COMPARISON

Javier Franke  
Janek Behrens  
Prof. Dr.-Ing. Stefan Gudenkauf

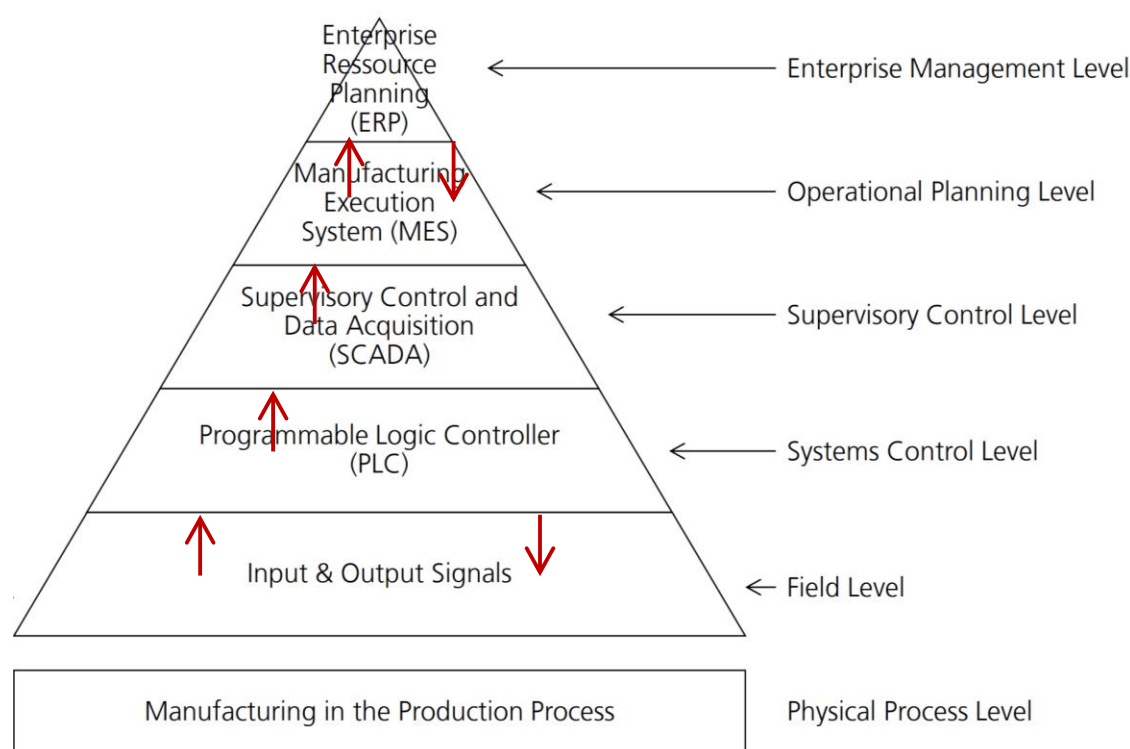
DigiWe 2023 | INFORMATIK 2023  
Jade University of Applied Sciences

# TABLE OF CONTENTS

1. Motivation
2. Problem Statement & Goal Definition
3. Methodology
4. Survey Completeness & Results Overview
5. EDAM-MQ Reference Model
6. Conclusions
7. Future Work

# FORCES TO THE PRODUCTION PROCESS MOTIVATION

Hierarchically organized



Point-to-point communication

High communication effort  
Limited flexibility due to rigid interfaces

IIoT and Industry 4.0

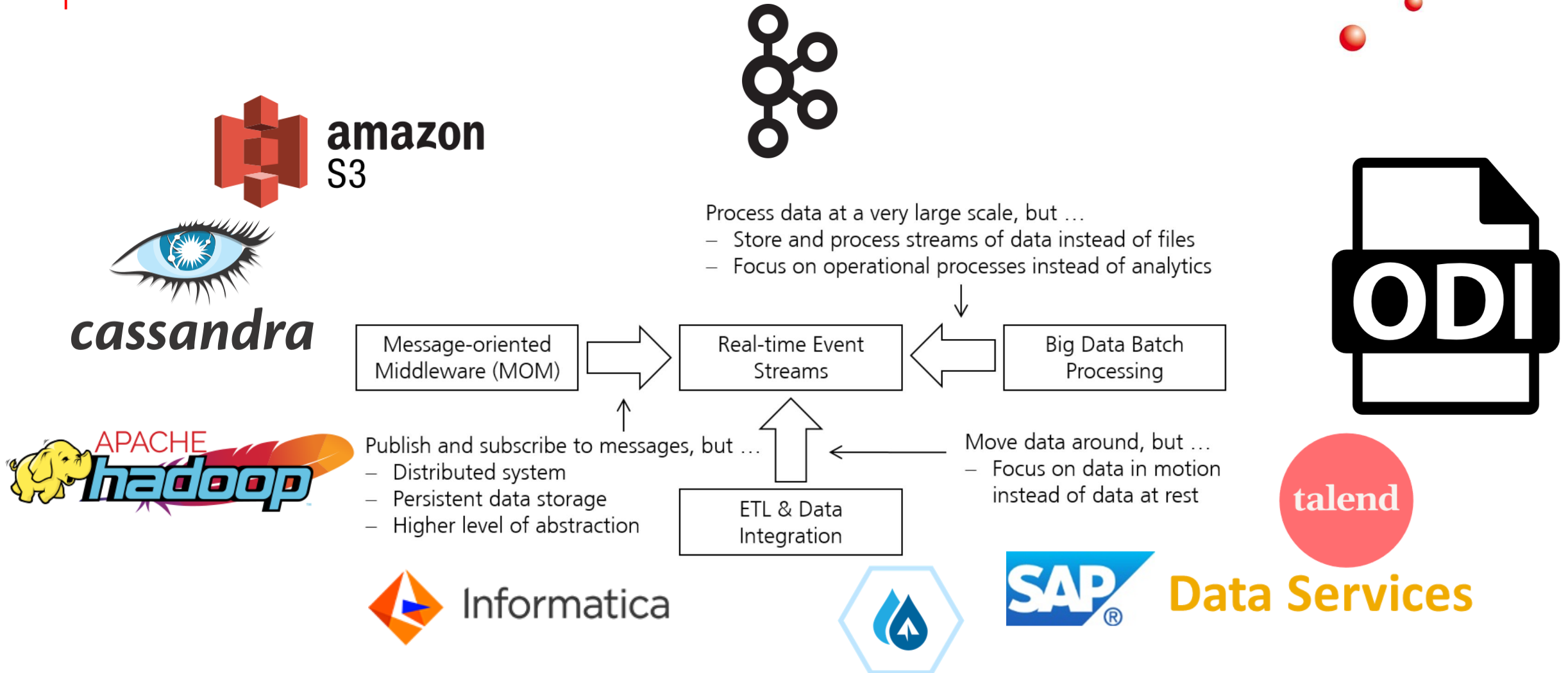
Servitization

Ever-increasing  
demand for  
optimization

Need to record, store and distribute  
data across all levels of the hierarchy

# EVENT-BASED SYSTEMS ARE ON THE RISE

## MOTIVATION



# PROBLEM STATEMENT & RESEARCH QUESTIONS

The ongoing trend towards service orientation of the entire automation hierarchy generates the opportunity to transfer the advantages of event-based software architectures to a holistic, data-streambased production in Industry 4.0 [SG16].

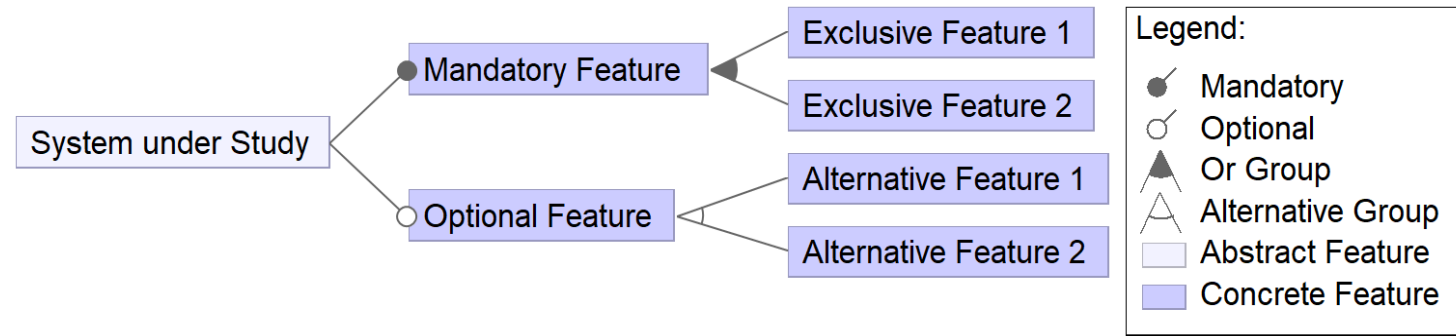
These architectures then play the role of event-based integration platforms or simply as IoT brokers.

However, potential users of such systems must find a way to uniformly describe and evaluate event-driven manufacturing systems, regardless of whether they want to formulate their own architectural needs or to evaluate the product offers of external providers.

**Goal:** Possibility of improving (purpose) the comparability (issue) of event-driven message-oriented middleware systems (object/process) from the viewpoint of the manufacturing industry (viewpoint).

1. What features do event-driven message-oriented middleware systems of the recent past have? → Survey
2. Can we provide a reference model, with which one can describe and compare these systems? → Feature Model

# METHODOLOGY



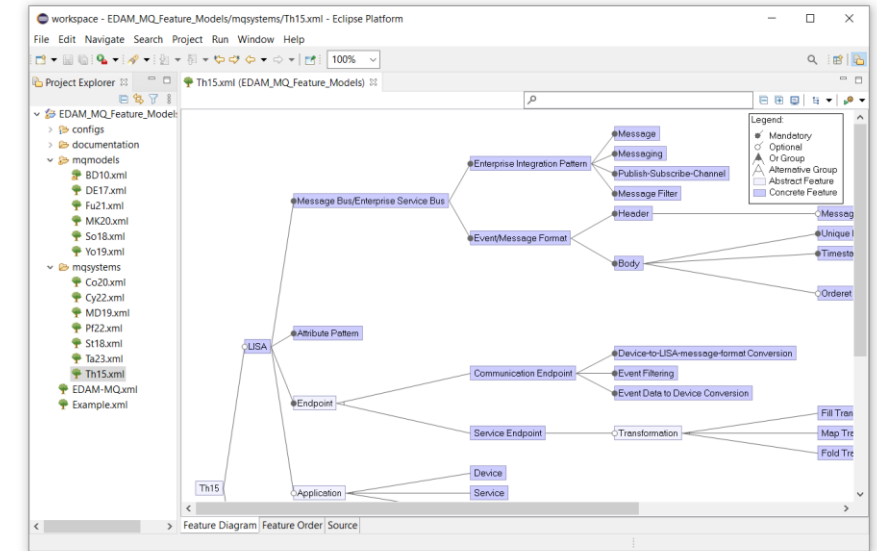
## Goal Definition: Goal Question Metric (GQM) Approach

### Survey:

- **Breadth-first** search using Google/Google Scholar, and SpringerLink
- **Keywords** included “event-driven”, “broker”, “message queuing” and “message-oriented” in conjunction with “manufacturing”
- **Selection criteria:** approaches that (1) are based on the Event-Driven Architecture (EDA) software architecture paradigm (2) especially based on Message Queuing (MQ) (3), generic or specifically oriented towards manufacturing/industrial production (4) works no older than 2015 (except for [BD10] as a reference)
- In-depth analysis by **feature modeling** [BA05], [CE00], [CH06] using FeatureIDE [Me17]

### Reference Model:

- Feature modeling [BA05], [CE00], [CH06] using FeatureIDE [Me17] (eliminating synonyms, uniform feature depth)



# SURVEY COMPLETENESS & RESULTS OVERVIEW

| Work   | Abstraction   | Domain        | Type of Work | Formality             | Provenance |
|--------|---------------|---------------|--------------|-----------------------|------------|
| [BD10] | model         | generic       | book         | informal              | academia   |
| [Fu21] | model         | generic       | paper        | informal              | academia   |
| [So18] | model         | manufacturing | paper        | informal              | academia   |
| [Yo19] | model         | generic       | paper        | informal              | academia   |
| [DE17] | model, system | generic       | paper        | informal              | academia   |
| [MK20] | model         | manufacturing | paper        | informal              | academia   |
| [St18] | system        | generic       | book         | informal              | industry   |
| [Sh21] | system        | generic       | book         | informal              | industry   |
| [Co20] | system        | generic       | whitepaper   | informal              | industry   |
| [Ta23] | system        | generic       | online       | informal              | industry   |
| [MS19] | system        | generic       | paper        | informal <sup>7</sup> | academia   |
| [Th15] | system        | manufacturing | paper        | informal              | academia   |
| [Cy22] | system        | manufacturing | whitepaper   | informal              | industry   |
| [Pf22] | system        | manufacturing | thesis       | informal              | academia   |

**Systems:** Apache Kafka [St18], Confluent Platform [Co20], Apache Pulsar [Ta23], RabbitMQ [MD19], LISA [Th15], Cybus Connectware [Cy22], and ZeroMQ [Pf22]

**Models:** Bruns and Dunkel [BD10], Fu et al. [Fu21], Sommer et al. [So18], Yongguao et al. [Yo19], Dobbelaere and Esmaili [DE17], and Mishra and Kertesz [MK20]

# EDAM-MQ REFERENCE MODEL RESULTS

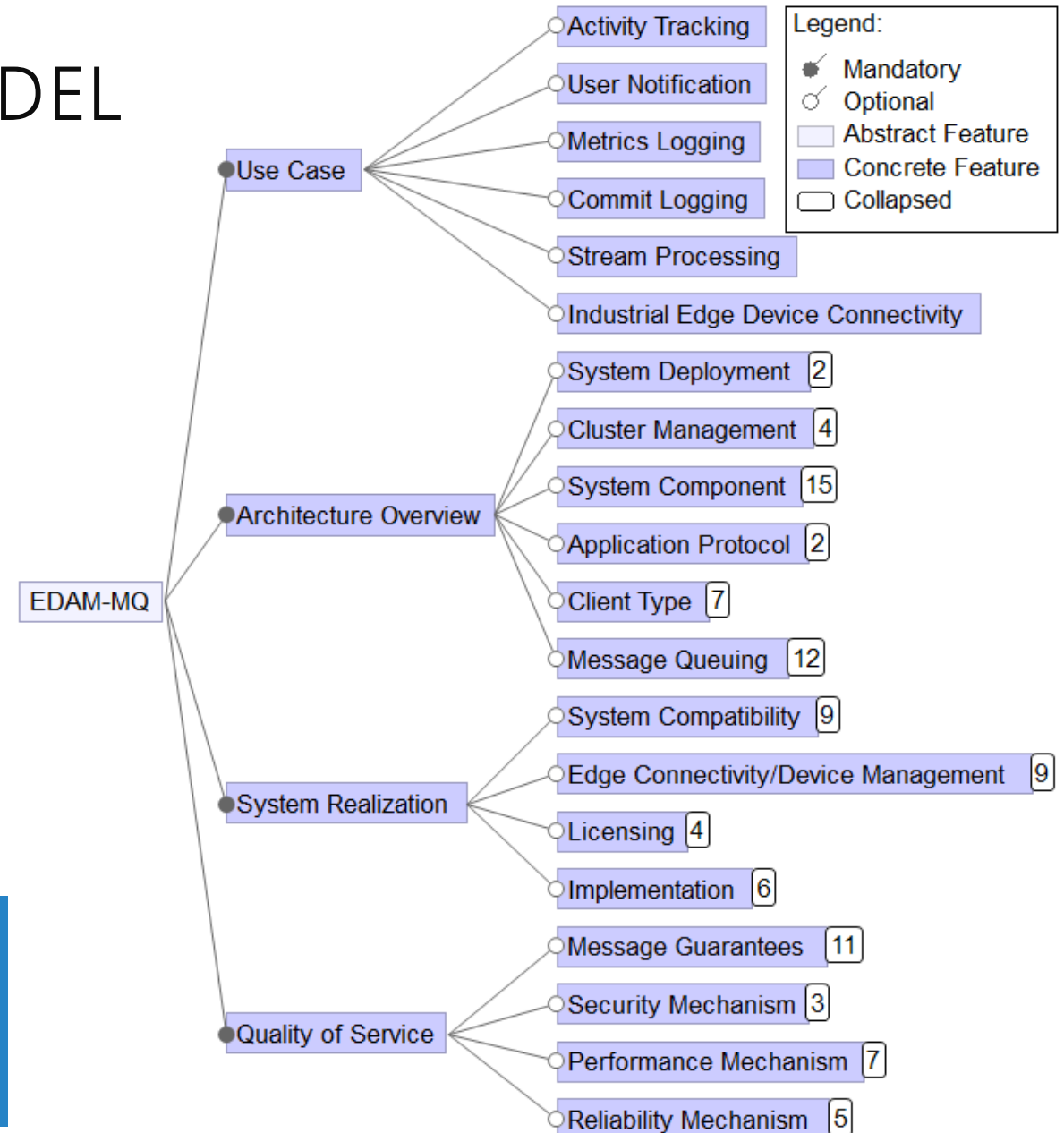
**Use Case:** Typical use cases for EDA in general and for production

**Architecture Overview:** Typical system architecture components

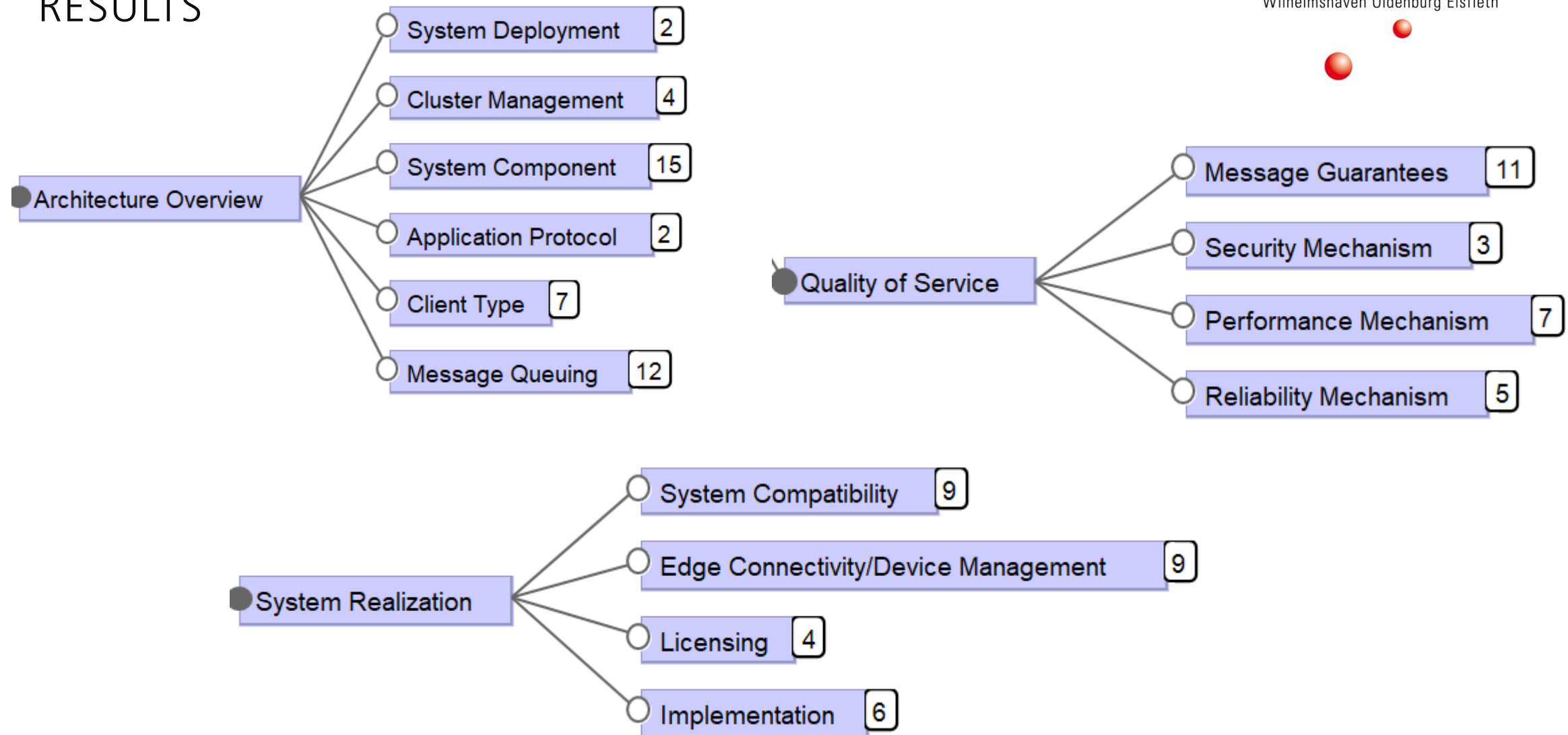
**System Realization:** Descriptions of the technical realization of the system

**Quality of Service:** QoS and Non-functional guarantees typically discussed

**Please note:** The selection state of a feature of an instance of the reference model (e.g. for Apache Kafka) denotes the presence (or absence) of the description of a system capability



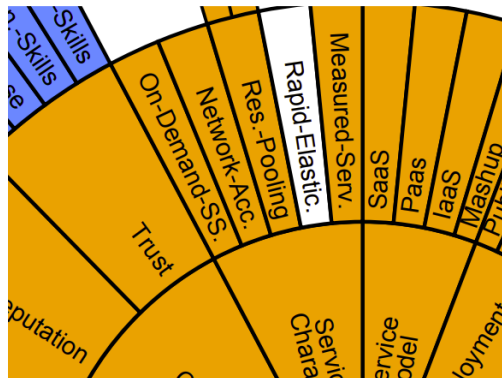
# EDAM-MQ REFERENCE MODEL RESULTS



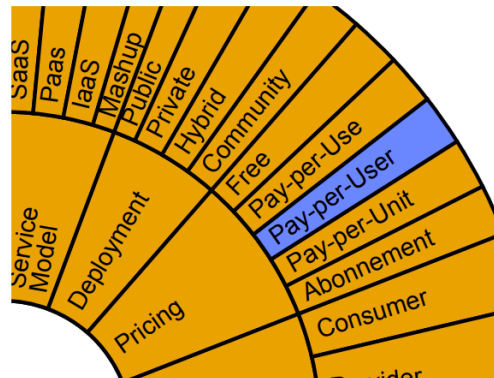
# HOW TO MAKE IT MORE ACCESSIBLE?

## FUTURE WORK

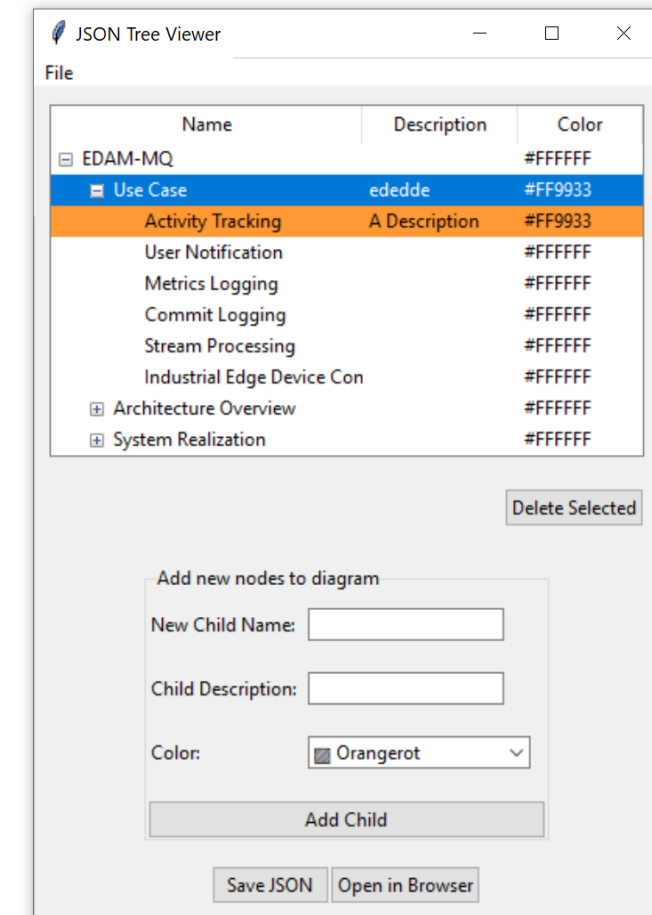
- Applying interactive visualization to feature models using interactive sunburst diagrams similar to the Service Navigators approach of [Gu13]
- Tool to import, annotate and visualize FeatureIDE XML files (to JSON and interactive plotly.py)
- Refine EDAM-MQ by conducting a structured literature review [Ki07] [KB13]
- Provide form-based input instead of feature modeling for EDAM-MQ



(a) Absent service feature.



(b) User-defined service feature.



# HOW TO MAKE IT MORE ACCESSIBLE?

## FUTURE WORK

- We surveyed recent models and architectures for event-driven message-oriented middleware systems
- Based on this survey, we propose a consolidated feature model EDAM-MQ to uniformly describe and evaluate event-driven manufacturing systems, regardless of whether an organization's own architectural needs or the offers of external providers are evaluated'.
- Additionally to the future work described on the previous slide, we envision the integration of interactive feature modeling for EDAM-MQ systems with Service Navigators (previous slide) and architecture analysis based on quality scenarios (e.g. by applying the Architecture Tradeoff Analysis Method) as a promising experiment to form a holistic decision system.

# DISCUSSION

Javier Franke  
Janek Behrens  
Prof. Dr.-Ing. Stefan Gudenkauf

DigiWe 2023 | INFORMATIK 2023  
Jade University of Applied Sciences

# REFERENCES

- [Ba94] Basili, V.; Caldiera, G.; Rombach, H. D.: Goal Question Metric Approach: Encyclopedia of Software Engineering. John Wiley & Sons, Inc., pp. 528-532, 1994.
- [Ba05] Batory, D.: Feature Models, Grammars, and Propositional Formulas. In (Hutchison, D. et al. Eds.): Software Product Lines. Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 7–20, 2005.
- [BD10] Bruns, R.; Dunkel, J.: Event-driven architecture. Softwarearchitektur für ereignisgesteuerte Geschäftsprozesse. Springer, Berlin, Heidelberg, 2010.
- [Bo08] Borthakur, D.: HDFS Architecture Guide. Hadoop apache project 53, 2008.
- [CE00] Czarnecki, K.; Eisenecker, U. W.: Generative programming. Methods, tools, and applications. Addison-Wesley, Boston, 2000.
- [Ce21] Cerquitelli, T. et al.: Manufacturing as a Data-Driven Practice: Methodologies, Technologies, and Tools. Proceedings of the IEEE 4/109, pp. 399–422, 2021.
- [CH06] Czarnecki, K.; Helsen, S.: Feature-based survey of model transformation approaches. IBM Systems Journal 3/45, pp. 621–645, 2006.
- [Co20] Confluent, Inc.: Confluent Platform Reference Architecture, 2020.
- [Cy22] Cybus: Cybus Connectware Technical Specification. <https://www.cybus.io/wp-content/uploads/2022/02/cybus-connectware-technical-specifications-siemens-industrial-edge.pdf>, accessed 22 May 2023.
- [DE17] Dobbelaere, P.; Esmaili, K. S.: Kafka versus RabbitMQ. The first public release of this report appeared in: P. Dobbelaere and K. S. Esmaili. Kafka versus RabbitMQ: A comparative Study of two Industry Reference Publish/Subscribe Implementations. In Proceedings of the 11th ACM International Conference on Distributed and Event-based Systems, pages 227–238. ACM, 2017. <http://arxiv.org/pdf/1709.00333v1>.

# REFERENCES

- [Di21] DIN EN IEC 62541-4:2021-08:2021-08-01, OPC Unified Architecture – Teil 4: Dienste (IEC 625414:2020); Englische Fassung EN IEC 625414:2020.
- [Ev04] Evans, E.: Domain-driven design. Tackling complexity in the heart of software. Addison-Wesley, Boston, Munich, 2004.
- [Fr21] Frochte, J.: Maschinelles Lernen. Grundlagen und Algorithmen in Python. Hanser, München, 2021.
- [Fu21] Fu, G.; Zhang, Y.; Yu, G.: A Fair Comparison of Message Queuing Systems. IEEE Access 9, pp. 421–432, 2021.
- [Gu13] Gudenkauf, S. et al.: A Reference Architecture for Cloud Service Offers. In (Gašević, D. et al. Eds.): Proceedings of the 17th IEEE International EDOC Conference. IEEE Computer Press, Vancouver, Canada, pp. 227–236, 2013.
- [Gu23] Gudenkauf, S.; Bachmann, U.; Hartmann, N.: A Concept and a Multitenant Web Application for Interactive Software Architecture Analysis. In (Sales, T. P. et al. Eds.): Enterprise Design, Operations, and Computing. EDOC 2022 Workshops. Springer International Publishing, Cham, pp. 268–283, 2023.
- [Jo21] Joseph, J.: Mastering Apache Pulsar. O'Reilly Media, Inc, 2021.
- [KB13] Kitchenham, B.; Brereton, P.: A systematic review of systematic review process research in software engineering. Information and Software Technology 12/55, pp. 2049–2075, 2013.
- [Ki07] Kitchenham, B. et al.: Guidelines for performing Systematic Literature Reviews in Software Engineering. Version 2.3. EBSE Technical Report, 2007.
- [KI17] Kleppmann, M.: Designing data-intensive applications. The big ideas behind reliable, scalable, and maintainable systems. O'Reilly, Beijing, Boston, Farnham, Sebastopol, Tokyo, 2017.

# REFERENCES

- [Kr11] Kreps, J.; Narkhede, N.; Rao, J.: Kafka: a Distributed Messaging System for Log Processing: NetDB'11, 2011.
- [La21] Langmann, R.: Vernetzte Systeme für die Automatisierung 4.0. Bussysteme - Industrial Ethernet - Mobile Kommunikation - Cyber-Physical Systems. Hanser, München, 2021.
- [LF22] Lewis, J.; Fowler, M.: Microservices. a definition of this new architectural term. <https://martinfowler.com/articles/microservices.html>, accessed 14 Jun 2022.
- [LM10] Lakshman, A.; Malik, P.: Cassandra: a decentralized structured storage system. ACM SIGOPS Operating Systems Review 2/44, pp. 35–40, 2010.
- [Me17] Meinicke, J. et al.: Mastering software variability with FeatureIDE. Springer, Cham, 2017.
- [Mi18] Mishra, B.: Performance Evaluation of MQTT Broker Servers. In (Gervasi, O. et al. Eds.): Computational Science and Its Applications – ICCSA 2018. Springer International Publishing, Cham, pp. 599–609, 2018.
- [MK20] Mishra, B.; Kertesz, A.: The Use of MQTT in M2M and IoT Systems: A Survey. IEEE Access 8, pp. 201071–201086, 2020.
- [MS19] Madhu, M. P.; Sunanda, D.: Distributing Messages Using Rabbitmq with Advanced Message Exchanges. International Journal of Research Studies in Computer Science and Engineering 2/6, 2019.
- [Nu21] Nuikka, J.: Comparison of Cloud Native messaging technologies. Master's thesis. Tampere University, 2021.
- [Oa14] OASIS Advanced Message Queuing Protocol (AMQP) TC: ISO/IEC 19464:2014(E). Information technology — Advanced Message Queuing Protocol (AMQP) v1.0 specification. ISO copyright office, Switzerland, 2014.
- [Pa08] Palankar, M. R. et al.: Amazon S3 for science grids. In (Kosar, T. Ed.): Proceedings of the 2008 international workshop on Data-aware distributed computing - DADC '08. ACM Press, New York, New York, USA, pp. 55–64, 2008.

# REFERENCES

- [Pf22] Pfefferkorn, D.: Verwendung von ZeroMQ in verteilter Automatisierung, Vorarlberg, 2022.
- [Ra19] Ramasamy, K.: Unifying Messaging, Queuing, Streaming and Light Weight Compute for Online Event Processing: Proceedings of the 13th ACM International Conference on Distributed and Event-based Systems. ACM, New York, NY, USA, p. 5, 2019.
- [SG16] Siepmann, D.; Graef, N.: Industrie 4.0 – Grundlagen und Gesamtzusammenhang. In (Roth, A. Ed.): Einführung und Umsetzung von Industrie 4.0. Grundlagen, Vorgehensmodell und Use Cases aus der Praxis. Springer Gabler, Berlin, Heidelberg, pp. 17–82, 2016.
- [Sh22] Shapira, G. et al.: Kafka. The Definitive Guide Real-Time Data and Stream Processing at Scale. O'Reilly, Sebastopol CA, 2022.
- [SJ14] Shahrivari, S.; Jalili, S.: Beyond Batch Processing: Towards Real-Time and Streaming Big Data. arXiv, 2014.
- [So18] Sommer, P. et al.: Message-oriented Middleware for Industrial Production Systems: 2018 IEEE 14th International Conference on Automation Science and Engineering (CASE). IEEE, pp. 1217–1223, 2018.
- [St18] Stopford, B.: Designing Event-Driven Systems. Concepts and Patterns for Streaming Services with Apache Kafka. O'Reilly, Sebastopol, 2018.
- [Ta23] The Apache Software Foundation: Apache Pulsar Docs Version 3.0.x. Concepts and Architecture. <https://pulsar.apache.org/docs/3.0.x/concepts-overview/>, accessed 28 May 2023.
- [Th15] Theorin, A. et al.: An Event-Driven Manufacturing Information System Architecture. IFAC-PapersOnLine 3/48, pp. 547–554, 2015.
- [Th14] Thüm, T. et al.: FeatureIDE: An extensible framework for feature-oriented software development. Science of Computer Programming 79, pp. 70–85, 2014.
- [Wo18] Wolff, E.: Microservices. Grundlagen flexibler Softwarearchitekturen. dpunkt.verlag, Heidelberg, 2018.

# REFERENCES

[Yo19] Yongguo, Jiang and Qiang, Liu and Changshuai, Qin and Jian, Su and Qianqian, Liu: Message-oriented Middleware: A Review: 2019 5th International Conference on Big Data Computing and Communications (BIGCOM). IEEE, pp. 88–97, 2019.

[Ze23] The ZeroMQ authors: ZeroMQ. An open-source universal messaging library. <https://zeromq.org/>, accessed 22 May 2023.